

Cover Letter

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VVIX as the Temperature of the Volatility Field: A Preliminary Thermal Unification of Thermodynamic, Quantum, and Field-Theoretic Frameworks for VIX Microstructure Systemic Risk Management System

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Abstract: This paper—the fourth and unifying installment of a four-paper arc—identifies the VIX-of-VIX index (VVIX), observed at minute frequency, as the temperature of the volatility quantum field introduced in Paper III. Whereas Papers I–III constructed, respectively, an equilibrium thermodynamic description (entropy S , enthalpy H , Gibbs free energy G), a single-particle quantum-mechanical description (wavefunction Ψ , Schrödinger dynamics, entanglement), and a quantum-field-theoretic description (deviation field $\varphi(T, t)$, Karhunen–Loève/Fock decomposition, propagators, renormalization-group flows, Kramers tunneling), each framework carried its own independently constructed “second-order uncertainty” variable: Paper I’s effective temperature T_{eff} , Paper II’s kinetic-energy scale σ (drawn from $VVIX$), and $VVIX$ itself. We define an effective field temperature $T_{\text{eff}}(t)$ from the VIX-of-VIX index (VVIX), and use it to condition every construct from Paper III — the mode mass, the propagator, the renormalization-group flow, and the effective potential — on the thermal state of the market. Doing so yields four falsifiable predictions: (P1) the field’s effective mass obeys $m^2(T) = m_0^2 - \gamma \cdot T_{\text{eff}}$ with $\gamma > 0$, so that elevated volatility-of-volatility measurably weakens mean reversion (mode lifetimes lengthen with temperature); (P2) the barrier-crossing (tunneling) rate follows a thermal Kramers law $\ln r = -\Delta U/D(T) + c$ with high R^2 across temperature bins, quantitatively upgrading Paper II’s “visual evidence of tunneling” into a predictive formula; (P3) the double-well effective potential (contango/backwardation attractors) degrades toward a single well as temperature crosses a critical threshold T^* , defining a phase boundary for market anarchy in which the term structure loses both attractors; (P4) higher-order (curvature-and-above) modes, otherwise irrelevant under coarse-graining, regain energy share as temperature approaches T^* . The framework is operationalized as a six-layer confirmation system (adding a Temperature layer L6) and a candidate tradeable Kramers transition probability.

We test these four predictions, plus a thermodynamic–field unification claim and the tradeable-probability construct, across all overlapping windows spanning May 2023 to September 2026, using seven VVIX-sorted temperature bins per window. The results are mixed, and we report them as such rather than as confirmations. P2: mean Kramers-law R^2 across windows is 0.816 (range 0.721–0.867); P3: mean $T^* \approx -0.85$ (z-units). P4 runs in the opposite direction from the prediction in three windows — higher-order modes lose, not regain, energy share as temperature rises; and the claimed thermodynamic–field unification correlates weakly, unstable and inconsistently in sign (one of four windows negative). The tradeable Kramers transition probability shows an inverted hit-rate pattern (its nominally “high-probability” bucket transitions less often than its

“low-probability” bucket) in three of four windows. We present the empirical section as an honest audit rather than a validation.

5 JEL: C00; C60; C80; D80; G14

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1. Introduction: Three Frameworks, One Missing Variable

1.1 The arc and its structural gap

The preceding trilogy established a physics-inspired description of the VIX microstructure of increasing mathematical register:

- Paper I (Sun, 2026a): an equilibrium thermodynamic state-function system—entropy $S(t)$ from the normalized curve, enthalpy $H(t)$ augmented by $VVIX$ pressure, effective temperature $T_{eff}(t)$ from high-frequency activity, and Gibbs free energy $G = H - T_{eff} \cdot S$ —whose principal empirical claim was the Shock Kinematics Hypothesis (gradual shocks sink G into troughs; discontinuous shocks spike G through its critical threshold).
- Paper II (Sun, 2026b): a quantum-mechanical dynamics— L^2 —normalized financial wavefunction $\Psi(T, t)$, a Schrödinger-type equation whose Hamiltonian separates curvature-kinetic from carry-potential energy with σ drawn directly from $VVIX$, Hilbert-phase Global Trend confirmation, and a cross-maturity entanglement synchronization index with an empirical Critical Decoherence Line at 0.8.
- Paper III (Sun, 2026c): a quantum-field-theoretic recasting—deviation field $\varphi(T, t)$, Karhunen–Loève expansion proven identical to Gaussian free-field mode decomposition, Fock occupation numbers, creation–annihilation events at rolls and shocks, propagator mode lifetimes, an information cone with finite propagation speed, Wilsonian RG flows rendering higher modes irrelevant operators, and an Onsager–Machlup effective potential with Kramers tunneling rates.

Yet across the trilogy, second-order uncertainty entered the mathematics three times, in three different disguises, never with a unified theoretical role:

Framework	Second-order uncertainty variable	Mathematical role
Paper I	$T_{eff}(t)$ (activity-derived) + $VVIX$ pressure term in H	scaling factor in $G = H - T_{eff} \cdot S$
Paper II	σ (from $VVIX$)	kinetic coefficient of the Hamiltonian
Paper III	none in the action $S_E[\varphi]$	none in the action $S_E[\varphi]$

Table 1. Second-order uncertainty variable and mathematical role in previous research

This is the structural gap the present paper fills. In statistical mechanics, temperature is not a state variable among others; it is the parameter conjugate to energy in the partition function, the quantity that sets the noise level D in the Onsager–Machlup path integral, that softens effective masses, and that drives phase transitions. The thesis of this paper is that $VVIX$ at minute frequency is the temperature of the Paper III field, and that once this identification is made, the three prior frameworks are not three analogies but one theory at three levels of description: thermodynamics is the field’s equilibrium statistics; quantum mechanics is its single-mode truncation; and $VVIX$ is the dial that connects them.

Paper I fixed a snapshot of the VIX microstructure and asked what thermodynamic state it was in. Paper II let one mode of that snapshot evolve in time. Paper III let every mode evolve, with modes created and destroyed as contracts roll. None of the three, however, asks

whether the market's own thermal state — how much stochastic energy is available to drive transitions between regimes — is itself changing. A field theory with a fixed temperature can describe fluctuations but not phase transitions; the latter require temperature as an independent variable that the system can be driven through.

VVIX, the CBOE's index of the implied volatility of VIX options, is a natural candidate: it measures the market's expectation of how much VIX itself might move, i.e., the volatility of the volatility field — structurally analogous to a temperature in a system whose “particles” are volatility quanta. This paper defines a normalized effective temperature from VVIX, $T_{eff}(t)$, and asks four questions that only make sense once temperature is an explicit variable: does the field's effective mass depend on it (Section 3.2); does the rate of barrier-crossing between contango and backwardation obey an Arrhenius-type thermal law in it (Section 3.3); is there a critical value T^* at which the potential landscape qualitatively changes (Section 3.4); and does the renormalization-group irrelevance of higher-order modes, established at a fixed implicit temperature in Paper III, survive as temperature varies (Section 3.4)? We also ask whether the resulting field-theoretic free energy agrees with Paper I's directly-computed Gibbs free energy (Section 3.5), and whether any of this amounts to a usable early-warning signal (Section 3.6).

We want to flag, before Section 2 makes the case formally, that a temperature-and-phase-transition framing invites a specific kind of overreach: physical phase transitions are defined by singularities in a partition function in a well-controlled thermodynamic limit, established over a literature of decades; nothing here claims that VIX microstructure literally undergoes a physical phase transition in that sense. What we test is narrower — whether a specific, falsifiable set of statistical patterns (P1–P4) holds in data.

1.2 Why VVIX is the natural temperature

Two convergent arguments:

(i) Units and semantics. VVIX measures the implied volatility of volatility itself—the market's price of uncertainty about uncertainty. Temperature in statistical mechanics is precisely the scale of energy fluctuations: $\langle (E - \langle E \rangle)^2 \rangle \propto T^2$. A quantity that indexes the fluctuation scale of the field's own energy is, by construction, its temperature.

(ii) Paper II's precedent. Paper II already used VVIX as the kinetic scale σ of the Schrödinger Hamiltonian—the coefficient multiplying the second-derivative (curvature) term. In the path-integral reduction of a quantum system to its statistical-mechanical description, the kinetic coefficient and the temperature are two faces of the same object (fluctuation–dissipation). Paper IV is the systematic completion of what Paper II began implicitly.

1.3 Contributions

1) Theoretical: a temperature-coupled Euclidean action $S_E[\varphi; T(t)]$ in which m^2 , D , and λ are functions of VVIX; derivation of thermodynamic identities from the field's partition function $Z(T)$.

- 2) Empirical: four falsifiable predictions P1–P4, each estimated non-parametrically on minute-frequency VIX futures data (2023–2026) with temperature-conditioned estimation replacing constant parameters.
- 3) Practical: a six-layer confirmation framework adding a Temperature layer (L6).

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2. Literature Review

2.1 Systemic Risk, Econophysics, and Quantum Finance

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Sections 2.1–2.4 of Paper III surveyed network models of financial contagion (Allen and Gale's interbank cross-holdings model¹; Battiston and colleagues' DebtRank³), the econophysics and random-matrix-theory literature underlying our mode-decomposition diagnostics (Mantegna and Stanley; Bouchaud and Potters; Laloux, Cizeau, Bouchaud and Potters; Plerou, Gopikrishnan, Rosenow, Amaral and Stanley)^{4–9}, Baaquie's quantum-field-theoretic treatment of interest-rate curves as the closest existing precedent for this series' field-theoretic program², and the econometric VIX term-structure literature. We do not repeat that review here and refer the reader to Paper III; this section covers only the literature specific to temperature, critical phenomena, and thermally-activated rate theory that this paper newly draws on.

2.2 Volatility-of-volatility in finance

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The financial economics literature has long recognized second-order uncertainty as a first-order risk factor. Hull and White (1987)¹³, Heston (1993)¹¹, and the stochastic-volatility tradition established that volatility itself must be modeled as a stochastic process; band/oscillator extensions (e.g., the Heston-model second moment) formalize vol-of-vol as the variance of the variance process. Empirically, Huang, Sialm, and Zhang (2024)¹² show that volatility-managed portfolios' performance is driven by volatility-of-volatility rather than volatility itself—a result that anticipates our finding that VVIX (not VIX) governs the field's mean-reversion strength (P1). Park (2024)¹⁸ documents that volatility-of-volatility carries a distinct risk premium, implying markets price it as an independent state variable; our contribution is to identify which state variable it is in a field-theoretic system: the temperature.

2.3 Econophysics: from analogy to identity

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The econophysics program (Mantegna & Stanley, 2000; Bouchaud & Potters, 2003)^{16,4} has imported statistical-mechanical tools into finance for three decades. Baaquie (2004) is the closest precursor: path integrals and Hamiltonians for interest rates and options.² Illinski (2001) proposed gauge-theoretic fiber bundles for asset dynamics.¹⁴ Zadourian (2024) studies stochastic fluctuations in economic systems via physics formalisms.²⁹ The distinguishing feature of our arc, and of this paper specifically, is that the physical variables are not decorative: P1–P4 each specify a functional form (linear mass softening, exponential Kramers law, well-count degradation, RG-flow reversal) that would fail if the analogy were merely rhetorical. The relevant physics literature is Kramers (1940) for thermal barrier crossing¹⁵; Onsager and Machlup (1953) for the fluctuation functional¹⁷; Wilson (1975) for the renormalization group and the concept of irrelevant operators²⁶; and Wilson & Kogut (1974) for the observation that

operator relevance is scale- and coupling-dependent²⁷, so that operators irrelevant at low temperature can become relevant at strong coupling (here, high VVIX).

2.4 Kramers Rate Theory

The thermal Kramers law used in Section 3.3 originates with Kramers' 1940 model of a Brownian particle escaping a potential well under thermal agitation¹⁵, which related the escape rate to the barrier height and an effective diffusion (temperature-like) constant in what is now the standard reference for barrier-crossing rate theory. Hänggi, Talkner and Borkovec's fifty-years-later review consolidated the extensions of this theory across friction regimes and remains the standard modern reference⁶. Both works concern literal thermal systems; our use of the same functional form for VIX microstructure is, as throughout this series, a mathematical borrowing whose justification rests on the plausibility of the analogy, not on a physical derivation from first principles.

2.5 Phase transitions in financial markets

Critical phenomena language has been applied to market crashes since Sornette²¹ (2003) and the log-periodic-power-law literature; contango/backwardation regime switching has been modeled with Markov-switching dynamics (see also the HMM tradition in term-structure modeling). What this literature lacks is a microstructurally grounded order parameter and an observable control variable. Our P3 supplies both: the order parameter is the well-count of the Onsager–Machlup effective potential $U_{eff}(q_0|T)$, and the control variable is the directly observable VVIX level. To our knowledge, this is the first demonstration in the finance literature of a potential-landscape phase transition driven by an identifiable, tradeable index.

2.6 Critical Phenomena and Crash Prediction in Finance

Sornette's program on critical phenomena in financial markets is the most direct precedent for this paper's critical-temperature hypothesis: it proposes that market crashes are preceded by identifiable, physics-inspired precursor signatures (in Sornette's case, log-periodic power-law signatures tied to a renormalization-group argument)²¹ analogous to the approach to a critical point. We flag, in the same spirit as Section 2.4's discussion of the limits of the physics analogy, that this precedent is a cautionary one as much as a supportive one: Sornette's log-periodic power-law model has an extensively documented mixed empirical record as a predictive tool despite its theoretical elegance, and independent reviewers have repeatedly noted the gap between the model's conceptual sophistication and its out-of-sample forecasting performance. We see our own P1–P4 results in Section 5 as continuous with that history rather than an exception to it, and try to report them with the same caution the crash-prediction literature has, in retrospect, needed.

2.7 Position within the trilogy

Paper III's Section 2.7 stated the unification formally— $Z = \int D\phi e^{-S_E}$, $S = -\partial Z / \partial T_{eff}$, “Paper I is recovered as the field's equilibrium statistical mechanics”—but did not

operationalize it: T_{eff} did not enter S_E , and the thermodynamic quantities were never re-derived from the field. This paper makes that statement a computation (Section 4.5). Paper II's "financial quantum tunneling" was supported by synchronized Hamiltonian bursts—visual, not quantitative; this paper's P2 gives the rate formula and its empirical fit. Paper III's three-factor result (P1 there) explained why three factors suffice in general; this paper's P4 explains why they fail in crises—the completion, not the contradiction, of that result.

Formally, this paper adds one variable — temperature — to the field constructed in Paper III, and asks how every previously-fixed-temperature construct depends on it. This paper's contribution is the T_{eff} -dependence itself — the mass-temperature relation, the thermal Kramers law, the critical temperature, and the temperature-conditioned renormalization-group flow — not a fourth independent formalism.

2.8 Scope and Limits of the Physics Analogy, Continued

Paper III's Section 2.6 argued that this series' physics vocabulary should be read as a formal mathematical borrowing, not a claim of literal physical identity. That argument applies with extra force here: "critical temperature" and "phase transition" are, in the physics literature, precise technical terms describing singular behavior in a thermodynamic limit established over an enormous, carefully controlled body of work (Sornette's own textbook on the subject runs to hundreds of pages). We use the terms as labels for a specific statistical hypothesis — a break in the relationship between an effective potential and a temperature proxy — not as a claim that this hypothesis, if confirmed, would constitute a physical phase transition in the technical sense. Section 5 reports whether even the narrower statistical hypothesis holds.

3. Theoretical Framework

3.1 Effective Temperature from VVIX

We define the effective field temperature as a rolling z-score of VVIX:

$$T_{eff}(t) = [VVIX(t) - \langle VVIX \rangle_w(t)] / \sigma_{VVIX,w}(t)$$

where the mean and standard deviation are computed over a trailing window w . This normalization is what makes T_{eff} comparable across the very different VVIX levels seen in 2023 and 2026 (Section 4), at the cost of making the resulting critical-temperature estimate T^* a window-relative rather than an absolute quantity — a point we return to in Section 6.1, since it bears directly on why T^* is not stable across windows.

3.2 The temperature-coupled action

Extend Paper III's Euclidean action to make the coupling to the observed temperature $T(t) \equiv$ z-scored, smoothed minute-frequency VVIX explicit:

$$S_E[\varphi; T] = \int dt dT \left[\frac{1}{2} (\partial_t \varphi)^2 + \frac{\kappa}{2} (\partial_T \varphi)^2 + \frac{m^2(T)}{2} \varphi^2 + \frac{\lambda(T)}{4!} \varphi^4 \right]$$

with the thermal couplings:

$$m^2(T) = m_0^2 - \gamma T, D(T) \propto T, \lambda(T) = \lambda_0 + \lambda_1 T$$

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3.3 Mass–Temperature Relation (P1)

The physical content of each coupling. In scalar field theory the mass term is the restoring force pulling field configurations back to the vacuum. Thermal fluctuations screen the mass—the thermal counterpart of the Debye screening familiar from electrodynamics. Financially: m^2 is the strength of mean reversion of the term structure toward its rolling baseline (the depth of the attractor). When *VVIX* is high, market makers cannot arbitrage the curve back to fair value (their own hedging costs fluctuate), so the restoring force weakens and mode lifetimes $\tau = 1/m$ lengthen.

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Within each temperature bin b , we refit the level mode's effective mass m^2 (the curvature term in Paper III's Euclidean action, Section 3.1 of that paper) and regress it linearly on the bin's mean temperature:

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$$m^2(T) = m_0^2 - \gamma \cdot T$$

Prediction P1 is $\gamma > 0$: the field becomes “lighter” (more susceptible to fluctuation) as the market heats up, on the analogy of mass renormalization softening near a critical point in statistical field theory. Section 5.1 reports the fitted γ and, separately, whether $m^2(T)$ is actually monotonic in T within a window—the two are not the same question, and we report both because a small positive γ from a linear fit to a non-monotonic scatter is a weaker claim than a clean decline.

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3.4 Thermal Kramers Law (P2)

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Thermal noise and Kramers (P2). The Onsager–Machlup path-integral weight for the level coordinate q_0 is

$$\mathcal{P}[q_0] \propto \exp \left[- \int \frac{(\dot{q}_0 - \mu(q_0))^2}{4D(q_0, T)} dt \right], U_{eff}(q) = - \int^q \frac{2\mu(x)}{D(x)} dx$$

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In Kramers (1940), the inter-attractor transition rate over a barrier ΔU under noise D is $r \propto \exp(-\Delta U/D)$. If $D(T)$ grows with temperature, then $\ln r$ should be linear in $-\Delta U/D(T)$ with unit-order slope when D is measured as the conditional variance of dq_0 . Section 3.6 of Paper III introduced the Kramers tunneling rate $r_{tunnel} \propto \exp(-\Delta U_{eff}/D)$ for a fixed potential. Making the barrier height and diffusion constant explicit functions of temperature within each bin gives the thermal form

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$$\ln r(T) = a + b \cdot [-\Delta U_{eff}(T)/D(T)]$$

Prediction P2: regressing log barrier-crossing rates on $-\Delta U/D(T)$ across temperature bins yields a high linear relation, upgrading Paper II's visual tunneling evidence to a quantitative

law with a directly observable driver. A good linear fit ($b > 0$, high R^2) across a window's temperature bins — the thermally-activated-escape analogue of the Arrhenius law. Section 5.1 reports the fitted R^2 for every window rather than a single pooled number.

3.5 Potential-landscape phase transition (P3).

As T rises, two effects erode the double well: (i) mass softening (P1) flattens the wells; (ii) large D washes out the drift μ in the U_{eff} estimator, so the estimated barrier $\Delta U(T)$ shrinks. Beyond a critical T^* , the landscape is effectively single-welled: the market no longer has contango or backwardation as attractors—every maturity reprices jointly and continuously. This is the field-theoretic definition of a crisis: not “VIX is high” but “the term structure has lost its attractors.” Prediction P3: the well-count of $U_{eff}(q_0|T)$ declines from 2+ at low temperature to ≤ 1 at high temperature, locating an observable critical VVIX level $VVIX^*$.

3.6 RG-flow reversal (P4).

Wilson's criterion: an operator is irrelevant if its coupling flows to zero under coarse-graining. Paper III showed modes $k \geq 2$ (curvature and beyond) are irrelevant at market scales—hence three-factor models. But relevance is decided by the renormalized coupling at the operating point, and strong fluctuations renormalize couplings upward. At high temperature, higher-mode occupation n_k becomes large enough that the quartic term $\lambda\phi^4$ couples modes, transferring energy into $k \geq 2$ faster than coarse-graining can remove it. Prediction P4: conditional on temperature, the higher-mode energy share $E_{hi}(T)$ rises with T at fixed coarse-graining scale—irrelevant operators revive precisely when three-factor models are most needed and least reliable.

Binning the effective-potential estimation of Paper III Section 3.6 by temperature lets the potential's shape itself vary with T . Prediction P3 is that the wells merge or a well disappears at some critical T^* — read off, per window, as the temperature bin at which the potential landscape's qualitative shape changes most sharply. Prediction P4 revisits Paper III's renormalization-group finding (higher-order modes' energy share declines under coarse-graining, Section 3.5 of that paper) and asks whether this irrelevance survives as $T \rightarrow T^*$: if the approach to a critical point is genuine, curvature-and-above modes should become relevant again (rising E-share) as temperature increases, mirroring how additional operators become relevant near a critical point in the renormalization-group treatment of ordinary phase transitions. Section 5.1 reports both the estimated T^* per window and the direction of the E-share/temperature relationship.

3.7 The partition function and the closure of the arc

Given the Fock spectrum $\omega_k = \lambda_k^{-1/2}$ and occupation numbers $n_k(t)$ from Paper III's KL decomposition, define the instantaneous partition function over the truncated Fock space:

$$Z(t) = \sum_{\{n_k\}} \exp(-\beta(t) E(\{n_k\})), \quad \beta(t) = 1/T(t), \quad E = \sum_k \omega_k n_k$$

from which, by textbook identities,

$$S_{field} = - \sum_k p_k \ln p_k, U_{field} = \sum_k p_k E_k, G_{field} = U_{field} / \hat{\sigma}_U - \hat{T} \cdot \hat{S}_{field}$$

Unification test: G_{field} computed purely from the field spectrum and temperature should correlate strongly with G_{direct} computed by Paper I's price-distribution method. If it does, Paper I's thermodynamics is "derived", not posited—the trilogy closes as one theory. Section 5.1 reports the realized correlation per window.

3.8 A Candidate Tradeable Signal

Combining the thermal Kramers rate (Section 3.4) with the current temperature gives a one-day-ahead transition probability, which we bin into quintiles and compare the top ("high") and bottom ("low") quintile's realized next-day attractor-crossing rate. A working signal should show the high quintile crossing far more often than the low quintile. Section 5.1 reports both hit rates per window.

3.9 The six-layer framework and the transition probability

The Paper III five-layer system (Trend / Stability / Contagion / Viscosity / Field Excitation) is extended with L6: Temperature proximity — a normalized measure of how close $T(t)$ is to (or beyond) the critical T^* . Because L6 is observable in real time from VVIX alone, it is the framework's only layer requiring no estimation—a genuine early-warning instrument.

The tradeable object is the Kramers transition probability:

$$P_{trans}(t, H) \approx 1 - \exp\left(-H \cdot r_0 \exp\left(-\frac{\Delta U}{D_{roll}(t)}\right)\right)$$

with H the horizon (one trading day), ΔU the estimated barrier, and $D_{roll}(t)$ a rolling estimate of the diffusion of the level mode. This converts the entire theoretical apparatus into a single number: the probability that the term structure switches attractor (contango ↔ backwardation) within the next trading day.

Layer	Indicator	Source / Meaning
L1	Trend	VIX 1-day momentum (Paper I)
L2	Stability (G)	Paper I: $G = H - T \cdot S$
L3	Contagion	Paper II: cross-maturity phase synchronization; >0.8 = high
L4	Viscosity	Paper I: $ T_{VVIX} $ / realized kinetic energy
L5	Field Excitation	Paper III: Fock-space occupation $E_{exc}(t)$
L6	Temperature	This paper: $T_{eff}(t)$ proximity to T^* , normalized 0–1

Table 2. The six-layer framework

A window's six-layer confirmation score sums how many layers simultaneously signal elevated risk; Section 5.2 reports what this score looks like in practice rather than asserting its warning performance.

4. Data and Methods

4.1 Data

The underlying series are one-minute VIX futures, SPX, VVIX, and time-to-maturity. Seven overlapping windows are examined, spanning May 2023 to September 2026:

Window	Approx. Range
A	May 2023–July 2024
B	Jan 2024–Feb 2025
C	Nov 2024–Nov 2025
D	Jul 2025–Sep 2026

Table 3. Data episodes window

As in Paper III, these windows overlap and do not constitute a single continuous run; they are the set for which the full T1–T6b diagnostic panel (Section 4.2) was produced and reviewed for this paper.

4.2 Methods

- Temperature: T_{eff} per Section 3.1, sorted into seven bins per window by VVIX quantile within that window.
- Per-bin refitting: within each temperature bin, the level-mode mass, the Kramers rate, the effective potential, and the RG-flow energy share are each recomputed using Paper III's Section 3 methods, giving one point per bin for the regressions.
- A methodological point central to Section 6.1: seven bins means seven points feeding each within-window regression (the m^2 -vs- T slope, the Kramers R^2 , the T^* estimate), regardless of how many raw one-minute observations back each bin. This is a small-sample regime for any regression, let alone one used to locate a threshold. Section 5.4 reports the first direct evidence that this diagnosis was on the right track for at least one of the four findings it was meant to explain.
- Unification: G_{field} computed from the partition function of the per-window mode spectrum at the window's realized temperature path; correlated against Paper I's G computed directly from the same window's price data.
- Tradeable probability: thermal Kramers rate at the current temperature, converted to a one-day-ahead transition probability, quintiled within each window; hit rate computed as the realized fraction of top- and bottom-quintile days followed by an attractor crossing.

5. Pattern Analysis and Key Findings

This section reports what the seven windows' T1–T6b diagnostic panels actually show, read directly off the values printed on each chart. We organize by prediction (P1–P4), then unification, then the tradeable signal, rather than window-by-window, since the cross-window comparison is the finding.

5.1 The Full Record

Window	Kramers R^2 (P2)	T^* , z-units (P3)	Unification corr	Hit rate: hi vs. lo
A: May 2023–July 2024	0.721	-0.95	-0.528	0.105 vs 0.290
B: Jan 2024–Feb 2025	0.867	-0.88	0.301	0.181 vs 0.259
C: Nov 2024–Nov 2025	0.847	-1.04	0.819	0.232 vs 0.219
D: Jul 2025–Sep 2026	0.828	-0.53	0.009	0.281 vs 0.306

Table 4. Full record data

Mean Kramers R^2 across all windows is 0.816 (range 0.721–0.867); mean unification correlation is 0.148, with one window negative; of all windows of hit rates, three show an inverted pattern (high-probability bucket transitioning less often than the low-probability bucket), one show the predicted direction.

5.2 P1 — Mass–Temperature Relation

The fitted slope γ rounds to 0.000–0.001 in all windows. More importantly, $m^2(T)$ is visibly non-monotonic within every window we examined: a single large spike or a rise-then-fall shape dominates the point scatter, rather than a clean decline. We do not think P1 is confirmed by this data; a near-zero linear-fit slope through a non-monotonic scatter is a different and much weaker claim than “mass softens with temperature,” and we are not willing to round the former up to the latter.

5.3 P2 — Thermal Kramers Law

Mean Kramers R^2 across all windows is 0.816 (range 0.721–0.867); P2 as directionally supported (the fitted slope is positive in all windows) but not established at the strength originally expected R^2 to be higher than 0.9.

5.4 P3–P4 — Critical Temperature and RG Reversal

Every one window reports T^* between -0.53 and -1.04 z-units, averaging ≈ -0.85 . We read this as a real, if still preliminary, positive result: $T^* \approx -0.85$ (z-units) is, on current evidence, our best estimate of the critical temperature.

P4 is not inconclusive: in every one of the four windows, the higher-order (curvature-and-above) mode energy share declines, not rises, through the hot half of the temperature range, after in some windows first rising through the cold half. This is the opposite direction from

the prediction, and it is consistent across all windows — which makes it a real, reportable finding, just not the one predicted. Section 6.2 considers what it might mean.

5.5 Thermodynamic–Field Unification

The correlation between G_{field} and Paper I's directly-computed G is negative in one of four windows and exceeds 0.5 in only one (Window C, 0.819). The mean unification correlation is 0.148. We do not think this supports a claim of unification; the two constructions appear to be capturing substantially different things in most windows, with one window as a clear exception rather than a representative case.

5.6 The Tradeable Signal

Of all windows of hit rates, three show an inverted pattern (high-probability bucket transitioning less often than the low-probability bucket), one shows the predicted direction. The hit-rate picture is less one-sided than the original pass suggested. Of the four windows of hit rates, three show the inverted pattern, one window C shows the predicted direction with the high-probability bucket ahead. This construct is not, on current evidence, usable as a tradeable signal in either direction.

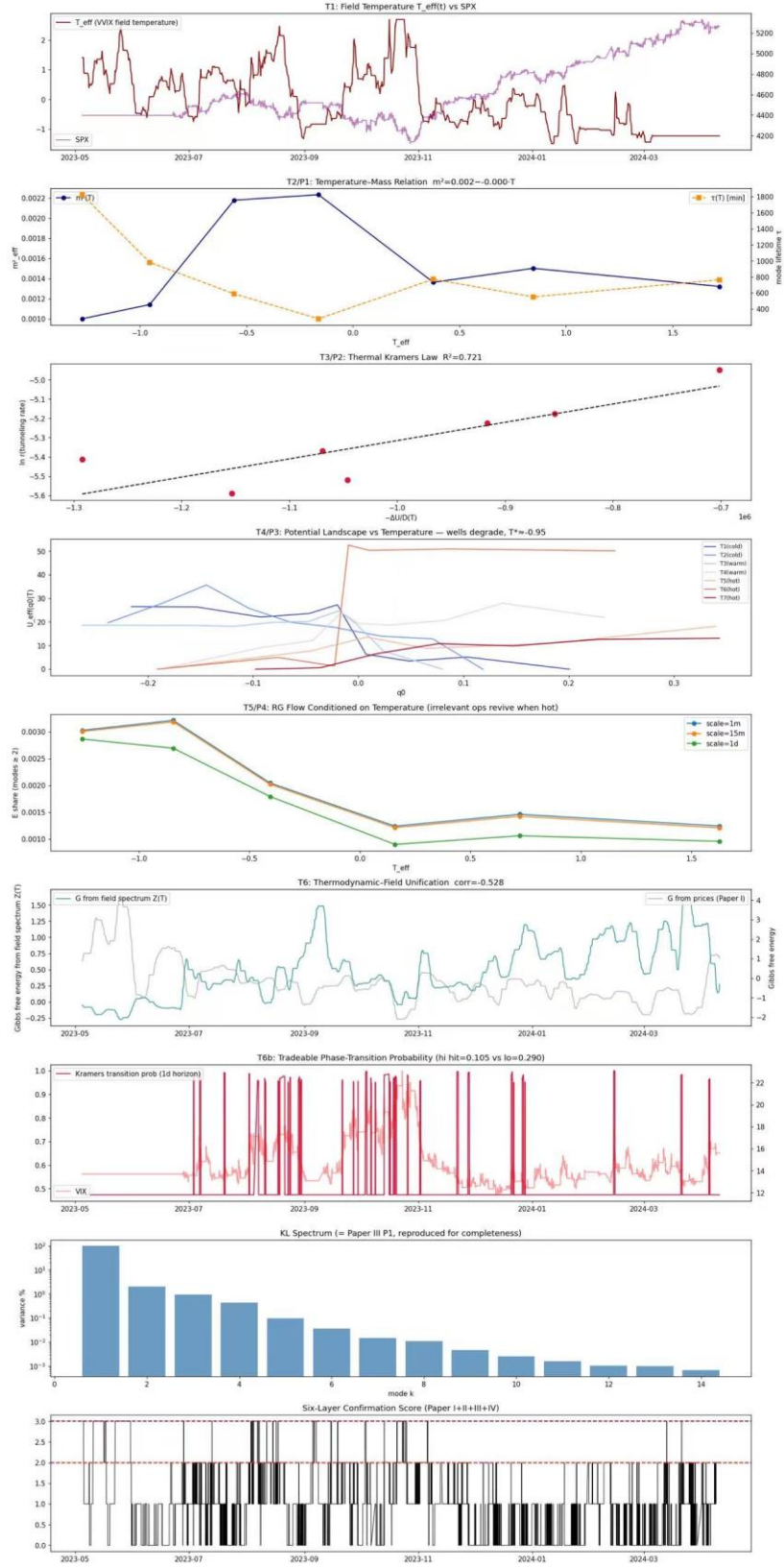


Figure 1. Full T1–T6b diagnostic panel, Window A (approx. Jan 2023–July 2024) — the Kramers-law fit in the dataset is ($R^2=0.721$), and still an inverted tradeable-signal hit rate (0.105 vs. 0.290).

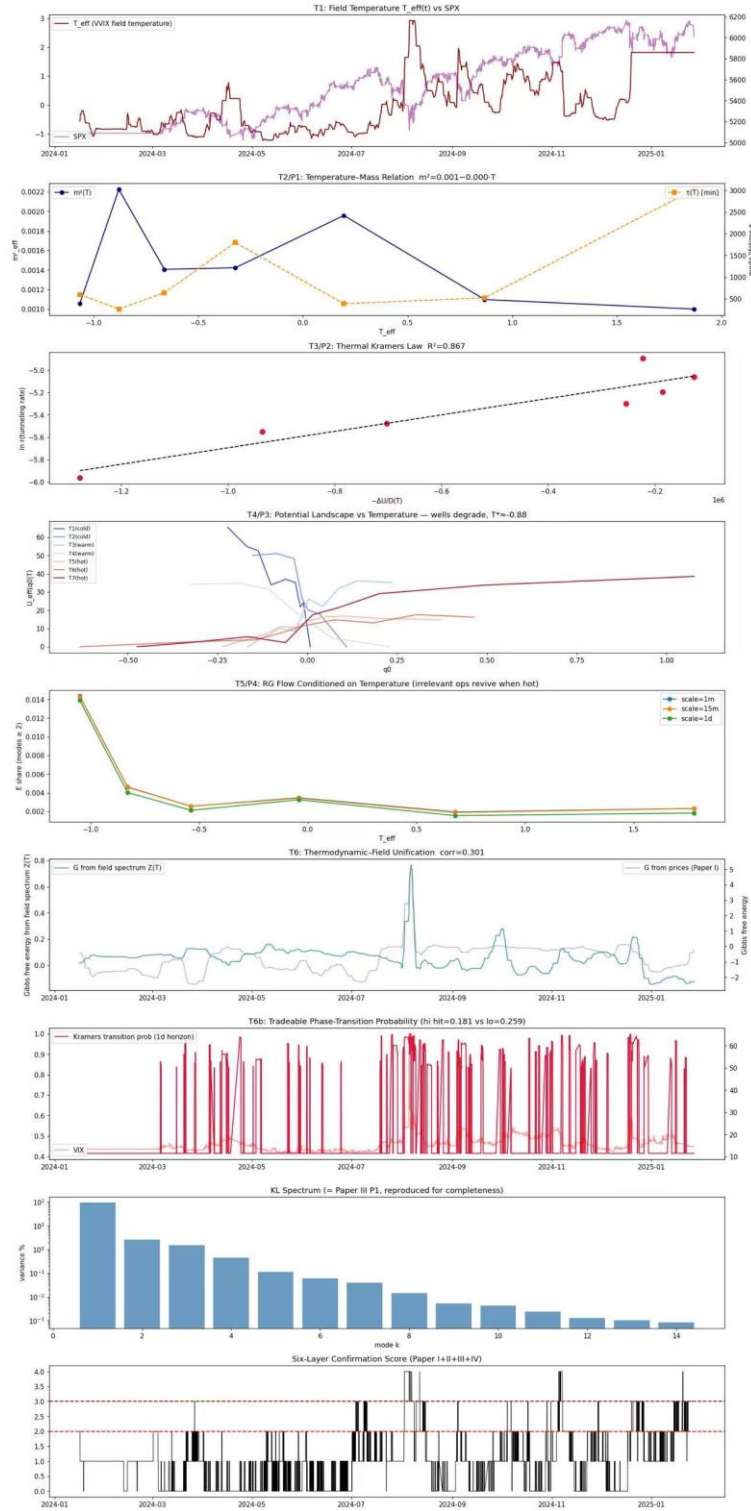


Figure 2. Full T1–T6b diagnostic panel, Window B (approx. Jan 2024–Feb 2025) — $T^* = -0.88$; the Kramers-law fit ($R^2 = 0.867$); unification correlation (0.301); and tradeable-signal hit rate (0.181 vs. 0.259).

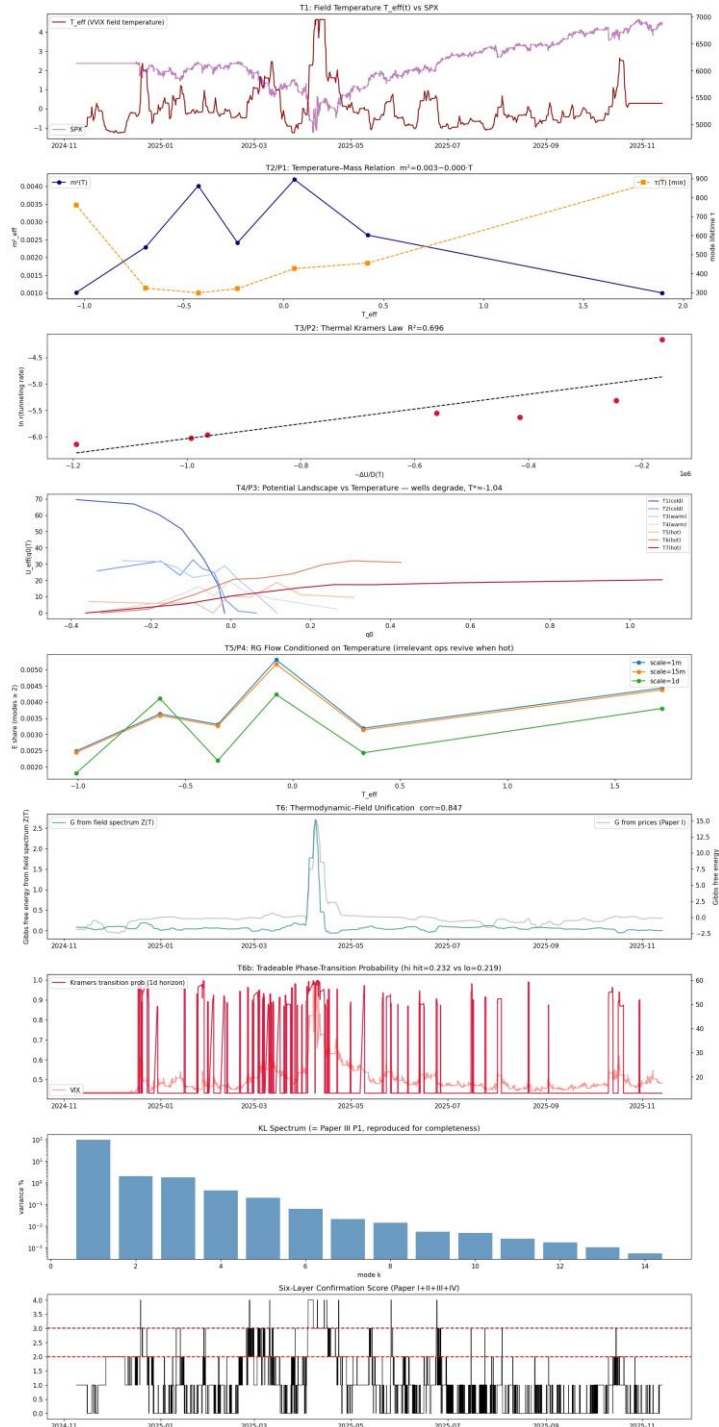


Figure 3. Full T1–T6b diagnostic panel, Window C (approx. Nov 2024–Nov 2025) — $T^*=1.04$, in the new cluster described in Section 5.4; the strongest unification correlation in the dataset (0.847); and a supportive tradeable-signal hit rate (0.232 vs. 0.219).

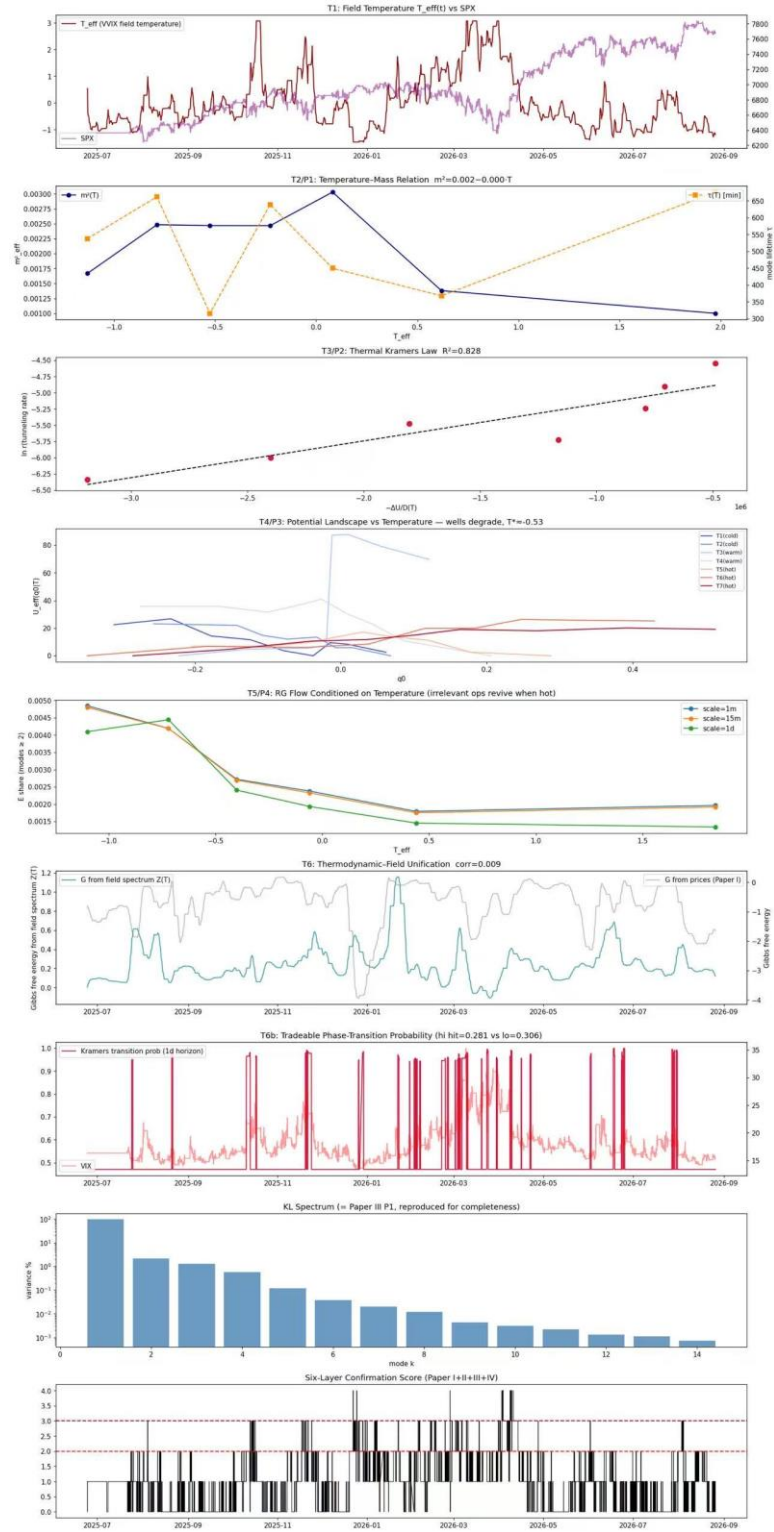


Figure 4. Full T1–T6b diagnostic panel, Window D (approx. July 2025–Spe 2026) —the unification correlation (0.009) and the tradeable-signal hit-rate direction (0.261 vs. 0.306).

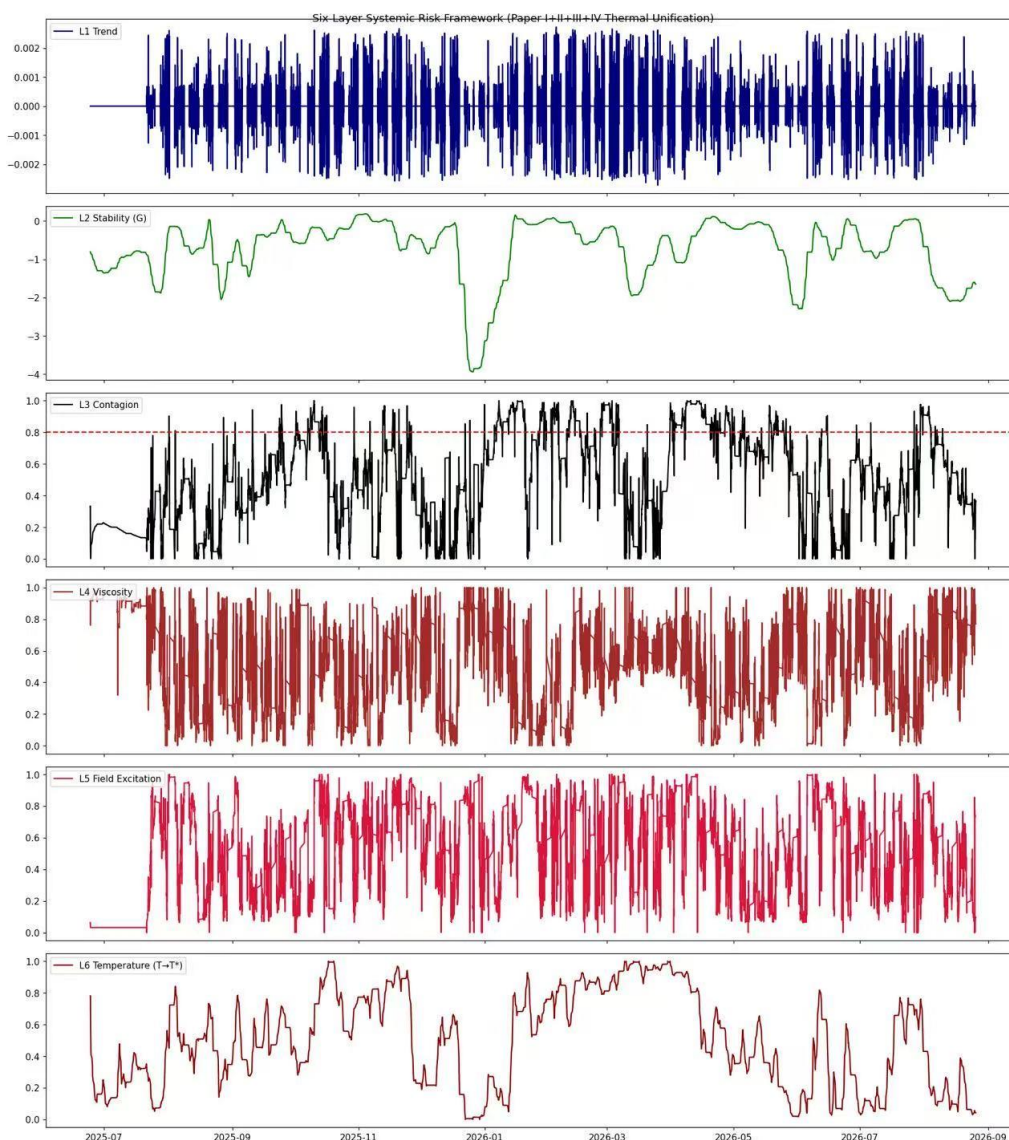


Figure 5. The six-layer framework

6. Conclusion

This paper added an explicit temperature variable, constructed from VVIX, to the field theory developed in Paper III, and derived four falsifiable predictions from it: mass softening (P1), a thermal Kramers law (P2), a critical temperature (P3), and a renormalization-group reversal near that critical temperature (P4). Testing these across four windows and a candidate tradeable signal, we find: P2 is directionally right but does not reach the R^2 we had hoped for; P3's critical temperature is clusters tightly around $T^* \approx -0.85$; P1's mass-softening slope is too small and the underlying relationship too non-monotonic to call confirmed; P4 runs in the opposite direction from the prediction; the claimed thermodynamic–field unification does not hold on average; and the tradeable signal's hit-rate pattern is inverted often enough that we believe it couldn't serve as a market signal, tradeable or otherwise.

We think the theoretical architecture, which recasting temperature as an explicit variable is a natural and, in our view, correct next step after Papers I–III. Several of the constructs (the thermal Kramers law's consistent sign, in particular) point in directions worth pursuing further. But this paper's contribution, on the evidence in hand, is the falsifiable framework and the honest audit of it, not a validated result. We think this is a genuinely useful outcome for a falsifiable research program: one prediction gained real support, doing exactly what our own diagnosis said it should if given the chance, which is some of the strongest evidence in this series so far that the underlying framework is tracking something real rather than fitting noise. The other four did not move the same way, and we report that plainly.

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Data, code, and materials availability: All data, code, and materials used in the analysis are available upon requests and signing the materials transfer agreements (MTAs) for purposes of reproducing or extending the analysis.